Paper Manager pack — 1st Place BEHAVIOR 2025 (Robot Learning Collective)

Paper: Task adaptation of Vision-Language-Action model: 1st Place Solution for the 2025 BEHAVIOR Challenge

arXiv: https://arxiv.org/abs/2512.06951 · PDF: https://arxiv.org/pdf/2512.06951.pdf

Code: https://github.com/IliaLarchenko/behavior-1k-solution

HF submission: https://huggingface.co/IliaLarchenko/behavior_submission (4 ckpts)

Session: Sep 9 · due 9:30 AM · **You = Empiricist (seat 1)**

Sources: Paper Summarizer + Senior Research (`DELIVERABLE.md`)

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1. Paper summary

Team **Robot Learning Collective** adapts **Pi0.5 / OpenPI** to the 50-task BEHAVIOR Challenge with:

- **Correlated-noise flow matching** (+ correlation-aware soft inpainting at infer)

- **Learnable mixed-layer KV attention**

- **System 2** stage tracking + voting (non-Markovian context for visual aliasing)

- **Multi-sample** FM training, **1.3×** cubic action compression

- Light **gripper / radio heuristics**

- Language dropped → **50 trainable task embeddings**; ship **4** group fine-tuned checkpoints

Result: 1st place — q_score **~26%** (pub **0.2605** / priv **0.2599**); binary success only **~11–12%**. Partial progress ≈ half of q-score. Authors call this a **competition report**, not a full ablation paper — opening for Empiricist.

Train cost (paper): 8×H200 FSDP; ~15 days multi-task + ~1 week/group FT; ~2 epochs; ~$13k. Infer: single **RTX 4090 24GB**.

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2. Important model / keywords

Kind	Terms
Backbone	Pi0.5, OpenPI, SigLIP-So400m/14, PaliGemma, action expert ~311M
Training	Flow matching, correlated noise β=0.5, multi-sample N=15, FAST aux λ=0.05, FSDP
Control / “planning”	Action chunk H=30, exec 26, soft inpaint 4, rolling RTC-style stitch, System 2 stages, gripper recovery
Data	10k demos, task embeddings (no NL), delta joints D=23, 30 Hz
Metrics	q_score (partial goal fraction), binary success, OmniGibson/BDDL
Deploy	4 FT ckpts + task→ckpt map; websocket `serve_b1k.py`

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3. Motion planning / long-horizon control (priority lens)

Not classical RRT/MPC. Stack = learned action-chunk FM + rolling receding-horizon-style execution + stage hierarchy + rule recovery.

Theme	Paper detail
Horizon	Avg episode ~6.6 min; longest ~14 min
Chunking	Predict **H=30** → execute **26** → overlap/inpaint **4**
Closed vs open	Mostly open-loop *inside* chunk; replan every 26 steps; soft inpaint (t>0.3) frees early steps for reactivity; **System 2 vote + gripper rules = closed-loop overrides**
Compression	Cubic 26→20 @ 30 Hz (**1.3×**); base vel ×1.3; disable on gripper change
Hierarchy	Stages 5–15/task; vote window 3 (≥2/3 forward; rollback rules)
Skill chaining	Implicit via stages — **no** skill library / separate planner passes (simpler than Hi Robot CoT)
Absent	Classical MP, MPC, ACT temporal ensemble, skill graphs

Empiricist talk spine: What fraction of “1st place VLA” is actually **outer-loop control** (gripper recovery + System 2), not train-time architecture?

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4. Role-ready bullets

Stakeholder (18+3) — seat 5

- **Motivation:** Multi-minute bimanual + mobile household chores still unsolved.

- **Problem:** 50 tasks; rank by partial-progress **q_score** on held-out instances.

- **Method:** Pi0.5 + correlated FM + System 2 + mixed-layer attn; multi-task IL; light heuristics; 4 specialized ckpts.

- **Findings:** ~26% q pub≈priv → 1st; binary ~11–12%; gripper heuristic huge on grasp subsets; multi-task → emergent recovery; undertrained (~2 epochs).

Scientific Reviewer (8+3) — seat 4

- **Strengths:** Code+HF; coherent correlated-noise train↔infer story; honest “not rigorous science”; pub≈priv; System 2 for aliasing.

- **Weaknesses:** No factorial ablations; task-ID embs + 4 ckpts weaken open-vocab VLA claim; heuristics mixed with science; underspecified LR/seeds; gripper 2.2× on only 13×3 eps.

- **Actionable:** Ablate β, System 2, inpaint modes, 1 vs 4 ckpts; seeds + curves; separate competition heuristics from methods tables.

Empiricist (11+3) — **YOU (seat 1)**

See §5–7 below. Headline hypothesis **H_emp** + P0 factorial E1×E2.

Archaeologist (11+3) — seat 2

- **Priors:** BEHAVIOR-1K; π_0/$\pi_{0.5}$; GR00T (last-layer XAttn foil); RT-1/2; FM / Diffusion Policy; FAST; ACT chunking; **RTC / real-time chunking**; Hi Robot (foil); SigLIP/PaliGemma.

- **Subsequent:** Too new (Dec 2025); author repo/blog/video only so far.

Visionary (11+3) — seat 3

- **2026 BEHAVIOR:** Bar still ~26% partial avg — room for recovery data, outer-loop System 2 / VLM checklists, unify to 1 multi-task ckpt, require with/without-heuristics reporting.

- **Follow-ups:** Corrective offline RL; privileged teacher; semantic stages; dexterity residual policies; real-robot correlated FM + soft inpaint vs RTC.

- Skip Discord/Drive logistics.

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5. Empiricist hypothesis (P0)

> **H_emp:** A large share of published q_score lift is **closed-loop recovery + non-Markovian stage context**, not only train-time architecture. Disabling the **gripper correction rule** drops partial success on

grasp-heavy tasks (order-compatible with paper's **~2.2×** subset claim); disabling **System 2 voting** raises order/confusion failures on visually ambiguous tasks; **both off** is worst.

Why Sep-9 feasible: Inference-only on HF submission ckpts — no 8×H200 month.

Success criteria: Directional gripper drop + System2 failure-mode shift. **Do not** gate on global q=0.26.

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6. Prioritized experiments (Senior Research)

Priority	ID	Experiment
P0	E1	Gripper-rule OFF
P0	E2	System 2 voting OFF
P0	E3	E1×E2 factorial ← **your talk figure**
P1	E4–E7	Soft inpaint / compression / exec horizon 10–26 / 1 multi-task ckpt vs 4
P2 / NOT Sep 9	E8–E12	Correlated noise retrain, attn freeze, multi-sample, FAST λ, full 50-task

Suggested subset: ~8–12 tasks (grasp-heavy + radio/microwave aliasing) × 3–5 instances × 4 conditions.

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7. Env setup + run scripts + GPU estimates

Outlines on box: `/workspace/vla-task-adapt-behavior/env_outline/`

(`environment.yml`, `setup_mamba_slurm.sh`, `sbatch_policy_server.sh`, `sbatch_eval_array.sh`, `run_ablation_matrix.sh`)

Setup sketch

```
# Clone solution + BEHAVIOR-1K; bash setup_remote.sh (upstream)# Prefer HF submission ckpts for P0 —
```

Policy server + eval

```
uv run scripts/serve_b1k.py --task-checkpoint-mapping task_checkpoint_mapping.json \  policy:checkpoi
```

Drive conditions via `run_ablation_matrix.sh` flags: gripper on/off, System2 on/off, optional compression / exec_horizon.

Train (P2 only — not required for your talk)

```
uv run scripts/train.py pi_behavior_b1k_fast \  --batch_size=2048 --num_train_steps=200000 \  --fsdp_
```

GPU / RAM budget (E1–E3)

Item	Estimate
Smoke (1 task × 1 inst × 2 cond)	**2–6 GPU-h**
Full P0 matrix (~10×4×3–5 eps)	**~40–120 GPU-h**
Policy VRAM	**~24 GB** (4090-class)
Eval node sys RAM	**≥48–64 GB** (leaks; don't pack many sims/node)

HF ckpt task splits (README)

- Ckpt1 (20): 2,3,5,6,10,11,13,14,15,19,23,24,25,28,29,34,42,44,47,48

- Ckpt2 (16): 0,1,7,8,9,12,16,17,18,20,21,22,26,30,43,45

- Ckpt3 (13): 4,27,31,32,33,35,36,37,38,39,41,46,49

- Ckpt4 (1): 40

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8. Slide Maker brief

Role	Timing	Spine
Stakeholder	18+3	Leaderboard + stack; **motion-control spine**: chunk H=30→exec26→inpaint4 + System2 + gripper
Reviewer	8+3	Strengths / missing factorials / heuristic leakage
Empiricist	**11+3**	**H_emp**, 2×2 factorial figure, env/run/GPU one-pager
Archaeologist	11+3	Lineage: ACT/RTC/$\pi_{0.5}$/BEHAVIOR (no Stakeholder redo)
Visionary	11+3	2026 bar + follow-ups (no Discord/Drive)

Style: Image-first; bullets in speaker notes. Prefer **PDF** export for Mac. Center long-horizon **control** story (not classical planner).

Quick paths

Artifact	Path
This pack	`/workspace/vla-task-adapt-behavior/KANTA_PACK.md`
Senior Research	`/workspace/vla-task-adapt-behavior/DELIVERABLE.md`
Env / sbatch	`/workspace/vla-task-adapt-behavior/env_outline/`
PDF paper	`/workspace/vla-task-adapt-behavior/paper.pdf`